

Ayush Maurya

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Mechanical Engineering with a minor in Mathematics

Sept 2021 – May 2025

Currently Enrolled in Graduate Certificate in Data Analytics Engineering

Sept 2025 – May 2026

Awards: Dean's List for the College of Engineering Spring 2022, Fall 2023, Fall 2024, and Spring 2025

Activities: Phi Gamma Delta, Northeastern Electric Racing, American Society of Mechanical Engineers (ASME)

Study Abroad: American College of Thessaloniki, Fall 2021

WORK EXPERIENCE

Draper Laboratory

Cambridge, MA

Electromechanical Engineering Co-op

Jan 2024 – Jun 2024

- Engineered 10+ CAD models and detailed technical drawings for high-precision inertial guidance sensor fixtures using PTC Creo, effectively implementing GD&T techniques to ensure machinability and tight tolerance control
- Created detailed assembly documentation and manufacturing drawings for complex electromechanical boxes, integrating wiring tables and routing constraints for technician assembly
- Performed an analytical thermal creep analysis in ANSYS to simulate epoxy mold deformation over extended durations, validating design performance under operational conditions and demonstrating understanding of material behavior
- Refined MATLAB simulations to determine safe particulate size thresholds for a capacitive sensor's working fluid, improving sensor reliability and noise performance

QinetiQ US

Waltham, MA

Mechanical Engineer Co-op

Jan 2023 – Jun 2023

- Collaborated with a cross-functional team of five engineers to design and procure parts for a large test fixture structure incorporating a sprocket drive, ensuring adherence to design requirements and quality standards
- Produced CNC-ready CAD models in SolidWorks/Creo with strict GD&T and DFM considerations, optimizing wall thickness, tool clearances, and tolerance stack-ups
- Redesigned a pan-tilt-zoom camera mount for a TALON robotic platform to resolve arm interference, iterating through rapid prototyping and hands-on assembly to deliver a collision-free, production-ready solution
- Utilized PTC Creo to develop the mount design, reflecting strong mechanical engineering principles and clear documentation for manufacturing directives
- Built and mechanically tested 3D-printed prototypes to validate design intent, characterize structural behavior, and accelerate design iteration
- Organized technical drawings, BOMs, and revision history in Windchill, ensuring full traceability and document control

ACADEMIC EXPERIENCE

Proton Therapy Mockup Capstone Project

Jul 2024 – Dec 2024

- Co-designed and manufactured a motorized, child-controlled proton therapy room mockup to reduce anxiety in pediatric cancer patients
- Applied FMEA and systems-level design thinking to ensure predictable motion, safe operation, and ease of maintenance
- Collaborated remotely with client representatives in Boston, New York, and the Netherlands

Northeastern Electric Racing

Aug 2022 – Dec 2023

- Analyzed brake rotor patterns to reduce thermal loading and fade during high-duty race conditions
- Designed structural mounts in SolidWorks for suspension system integration, prioritizing stiffness, mass efficiency, and manufacturability

Wheelchair Seat Warmer

Feb 2022 – Apr 2022

- Worked with three others to create a circuit with Arduino motors to transfer rotational energy to thermal energy
- Designed gears on AutoCAD and SolidWorks which rotated stepper motors as the wheels of the chair spun
- Laser-cut gears and 3D-printed casings to house two motors and gears which were attached to the rods of the wheelchair

COMPUTER & MACHINING SKILLS

Mechanical: SolidWorks, PTC Creo, AutoCAD, ABAQUS, ANSYS, MATLAB, Arduino, Python

Machining: Proficient with milling machine and bandsaw, basic knowledge of lathe